# Sungmin Yun

Computer Architectures and Systems for AI

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## About Me

I am a computer architecture and systems researcher at Seoul National University, focusing on **AI acceleration**. I have extensive hands-on experience in **accelerator design, hardware profiling, kernel programming, and performance modeling**, with strong expertise in **modern AI architectures** (e.g., MoE, MLA, RAG, diffusion). I lead research to **maximize hardware utilization** for LLM serving and to accelerate **Multi-LoRA inference** and **RAG pipelines**. I recently analyzed **hardware bottlenecks** in modern LLMs and derived serving guidelines, summarized in an arXiv preprint. I also developed an **open-source end-to-end LLM performance simulator**, LLMSimulator, enabling rapid evaluation of emerging hardware and model architectures. My research aims to **reduce the cost of AI** and make advanced models broadly accessible.

## Selected Publications

[MICRO 2024] Duplex: A Device for Large Language Models with Mixture of Experts, Grouped Query Attention, and Continuous Batching, **S. Yun**, K. Kyung, J. Cho, J. Choi, J. Kim, B. Kim, S. Lee, K. Sohn, J. H. Ahn, (Acceptance Rate: **22**% (113 among 497))

[ICS 2024] CLAY: CXL-based Scalable NDP Architecture Accelerating Embedding Layers, **S. Yun**, H. Nam, K. Kyung, J. Park, B. Kim, Y. Kwon, E. Lee, J. H. Ahn, (Acceptance Rate: **36**% (45 among 125))

[IEEE TC 2023] GraNDe: Efficient Near-Data Processing Architecture for Graph Neural Networks, **S. Yun**, H. Nam, J. Park, B. Kim, J. H. Ahn, E. Lee, (**Best of CAL Award**)

[arXiv 2025] The New LLM Bottleneck: A Systems Perspective on Latent Attention and Mixture-of-Experts, **S. Yun**, S. Park, H. Nam, Y. Lee, G. Lee, K. Kyung, S. Kim, N. S. Kim, J. Kim, H. Kim, J. Cho, S. Baek, J. H. Ahn

## Experience

**Postdoctoral Researcher**, Seoul National University — Sep. 2025 – Present

Lead the following research projects (3 manuscripts under review):

- **LLM Serving Systems** – develop system-level guidelines and optimizations for LLMs (e.g., MLA, MoE)
- **Multi-LoRA Inference at Scale** – lead end-to-end accelerator design and implementation
- **RAG Pipeline Acceleration** – co-design search algorithms and hardware for large-scale RAG pipelines

**Internship**, Samsung Electronics — Mar 2023 – Aug 2023

- Designed and implemented an RTL controller for general matrix–vector multiplication using HBM-PIM
- Analyzed architectural limitations of HBM-PIM and proposed Duplex, an LLM accelerator integrating custom HBMs with xPUs (later published at MICRO 2024)

## Education

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|---|---|---|
| **Seoul National University**, Ph.D. in Artificial Intelligence | GPA: 4.1/4.3 | Aug 2020 – Aug 2025 |
| **Yonsei University**, B.S. in Integrated Information Technology | GPA: 3.6/4.3 | Mar 2017 – Feb 2020 |

## Technical Skills

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| **Programming languages** | C/C++, CUDA, Python, System Verilog |
| **ML/LLM Frameworks** | Pytorch, vLLM, SGLang, TensorRT-LLM |
| **Profiling, Analysis, Simulators** | Nsight Compute, Nsight Systems, VTune, Ramulator |

## Honors & Awards

**Best Ph.D. Dissertation Award** Aug 2025
Seoul National University

**Best of CAL Award** Feb 2023
29th IEEE International Symposium on High-Performance Computer Architecture

## Invited Presentations

Poster presentation at Samsung AI Forum Nov 2023

Oral presentation on Best of CAL session at IEEE International Symposium on High-Performance Computer Architecture Feb 2023

## Program Committee and Paper Review

Lightweight Program Committee Member, ISCA 2026

IEEE Transactions on Computers

IEEE Transactions on Circuits and Systems for Artificial Intelligence

IEEE Computer Architecture Letters

Future Generation Computer Systems

## Additional Publications

[IEEE CAL 2025] SSD Offloading for LLM Mixture-of-Experts Weights Considered Harmful in Energy Efficiency, K. Kyung, **S. Yun**, J. H. Ahn

[IEEE CAL 2025] COSMOS: A CXL-Based Full In-Memory System for Approximate Nearest Neighbor Search, S. Ko, H. Shim, W. Doh, **S. Yun**, J. So, Y. Kwon, S. Park, S. Roh, M. Yoon, T. Song, J. H. Ahn

[HPCA 2025] Anaheim: Architecture and Algorithms for Processing Fully Homomorphic Encryption in Memory, J. Kim, **S. Yun**, H. Ji, W. Choi, S. Kim, J. H. Ahn

[IEEE CAL 2022] GraNDe: Near-Data Processing Architecture With Adaptive Matrix Mapping for Graph Convolutional Networks, **S. Yun**, B. Kim, J. Park, H. Nam, J. H. Ahn, E. Lee

[MICRO 2021] TRiM: Enhancing Processor-Memory Interfaces with Scalable Tensor Reduction in Memory, B. Kim, J. Park, **S. Yun**, E. Lee, M. Rhu, J. H. Ahn

## Patents

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| • Heterogeneous Computing Memory Device and Artificial Intelligence Computing System Including the Same | US<br>KR (1020250050877) |
| • Memory Device Capable of Dynamically Switching Between Error Correction Code and Error Detection Code in On-Die Error Correction Code | KR Granted (10-279756)<br>PCT (PCT/KR2021/019871) |
| • Memory Module and Computing System Including the Same | US (19/017,235)<br>KR (1020240060980)<br>CN (202510116255.2) |
| • Memory Controller and Operation Method Thereof | US (19/002,383)<br>KR (1020240050775)<br>CN (202411903289.8 |
| • Method and Apparatus for Memory Device Including Processing-in-Memory (PIM) | US<br>KR (1020250026131) |
| • Memory System and Operating Method Thereof | KR (1020250195434 |